

Multi-Aspect Multi-Sentiment Analysis with Attention

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Abstract

Aspect-based sentiment analysis (ABSA) involves the identification of the polarity (e.g., positive, negative, neutral) of a sentence given an aspect within that sentence. Our aim is to formulate several experimental hypotheses about such a type of sentiment analysis task and propose methods for their evaluation. To this aim, our main practical contribution is to report the results of experimentation with various neural network architectures that incorporate the aspect, and the application of an attention mechanism to the aspect, in different ways. Given our aim, our principle goal is to determine the effects of these differing network architectures as opposed to achieving strictly maximal performance. Through conducting our experiments on a challenge dataset we found that, absent the aspect, classification accuracy is worse than random; that the application of an attention mechanism significantly improves classification performance; and that an aspect-integrated attention approach does not necessarily improve upon a simple attention approach.

Introduction

Sentiment analysis and, in particular, aspect-based sentiment analysis has been of great importance to the field of Natural Language Processing for many years. Interest in the field has been amplified by recent advances in the application of deep neural networks to various sentiment analysis tasks. Many deep learning methods have been developed that have displaced previously state-of-the-art approaches. The result of these advances is that deep neural networks now dominate the sentiment analysis field. However, challenges persist [1]. Within the field, the difficulty/complexity of the aspect-based sentiment analysis task is magnified by the inclusion of multiple aspects within the same sentence. Consequently, different aspects will likely have different polarities. For instance, the sentence “*The decor was unsightly but the food was magnificent.*”, contains a negative polarity related to the aspect *decor*, whilst the *food* aspect has a positive polarity. This sub-type of ABSA is known as Multi-Aspect Multi-Sentiment (MAMS) and was introduced by Jiang et al. in the paper, *A Challenge Dataset and Effective Models for Aspect-Based Sentiment Analysis* [2].

Previous to the introduction of the MAMS dataset, the context of the typical ABSA task involved the classification of sentiment polarity of a given sentence that contained a single aspect or multiple aspects with the *same* polarity. The consequence of this scenario, as outlined by Jiang et al., is that the ABSA task degenerates to sentence-level sentiment analysis - making it difficult to determine the effect of various classification approaches when a sentence can contain multiple *different* aspects with multiple *different* polarities. The difficulty arises due to the empirical observation that sentence-level sentiment classifiers are able to produce high relative levels of accuracy without considering the aspect and will therefore fail to achieve high relative accuracy when evaluated on the MAMS dataset [2].

With these considerations in mind, we propose to evaluate several experimental hypotheses that seek to determine the effect of the integration of the aspect and the application of two different types of attention mechanism that incorporate the aspect. To this end, we seek to confirm or disconfirm the following experimental hypotheses:

- **Experimental Hypothesis 1:** Integration of aspect information will improve performance over a baseline Null Model that is *unaware* of the aspect.
- **Experimental Hypothesis 2:** The incorporation of a *simple attention* mechanism to an aspect integrated model will improve classification performance.
- **Experimental Hypothesis 3:** A model with an aspect integrated attention mechanism, i.e., *aspect attention*, will improve classification performance over the *simple attention* model.

To test these hypotheses we design a bi-directional recurrent Long Short Term Memory (LSTM) neural network model that is configurable to the requirements of each of the stated experimental hypotheses. Specifically, to test Experimental Hypothesis 1 (EH1) we utilize an LSTM that does not incorporate the aspect term, i.e., the model is *unaware* of the aspect. This model (Model 0) is the Null Model which we use as a comparative baseline. We compare this Null Model against an LSTM of the same configuration but which integrates the aspect term (Model 1). For Experimental Hypothesis 2 (EH2) we evaluate Model 1 against a model of the same configuration with the addition of a *simple attention* (i.e., dot-product attention) mechanism (Model 2). Finally, in order to evaluate Experimental Hypothesis 3 (EH3) we compare Model 2 with a model that utilizes a more involved attention mechanism, i.e., *aspect attention* (Model 3) - the architecture of which is inspired by a model design proposed by Wang et al., in *Attention-based LSTM for Aspect-level Sentiment Classification* from *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing* [3]. See the Appendix for a Model description reference.

Our main findings that result from testing our experimental hypotheses are, firstly, that the integration of aspect information significantly improves classification performance over a baseline Null Model. Secondly, the incorporation of a *simple attention* mechanism to an aspect integrated model additionally improves classification performance. Thirdly, a model with an aspect integrated attention mechanism, i.e., *aspect attention*, does not necessarily lead to improved classification performance over the *simple attention* model. In summary, we confirm both EH1 and EH2, but do not confirm EH3. This final result - non confirmation of EH3 - was an unexpected finding.

Methods

The purpose of the LSTM neural network under consideration is to test the experimental hypotheses outlined in the previous section. Consequently, the configuration of the network architecture relates directly to each experimental hypothesis. Illustrated in Figure.1 is the flow of the input sentence and aspect pair through the network. The route through the network that the input sentence and aspect pair take is dependent on the experimental hypothesis being tested. At a high level, the input sentence and aspect are transformed into an embedding space before flowing through the LSTM and into the attention function (left of Figure 1) or through the LSTM without attention (right of Figure 1). The attention function operates on the hidden-state and the LSTM output (i.e., *simple attention*) or on the hidden-state, the LSTM output and the aspect embedding (i.e., *aspect attention*). The outputs (i.e., hidden-state/context) are linearized before polarity, $\{positive, negative, neutral\}$, probabilities are deduced via the *softmax* function (see Appendix). Formally, given a sentence S and aspect term a , the neural network model is a function $f(S, a) = p$, where $p \in \{positive, negative, neutral\}$ is the polarity of the sentence given the aspect. We now describe in turn each of the functional components of the model architecture.

In the first layer of the model the input sentence S and aspect a are transformed into learned embedding representations. With a sentence of length n as a vector of words, i.e., $S = \{w_1, w_2, \dots, w_n\}$, the sentence embedding is a vector transform of S into the embedding space E^d where d is the dimension of the embedding space. Similarly, the aspect a is a word vector of length $n = 1$ and is transformed into the embedding space in the same way. We produce the intermediate aspect *aware* sentence representation via concatenation of the sentence and aspect representations.

The LSTM layer is composed of a sequence of functional units known as cells. Graphically, a bi-directional LSTM is illustrated, in unrolled form, by sequential layers atop one-another with opposing directions, i.e., forward and backward (left of Figure 1). Alternatively, in recursive form, the depiction is with a feedback

loop (right of Figure 1). Each cell of an LSTM contains within it an input i , forget f , and output gate o , which operate on the input X . We describe the functions as follows:

$$X = \begin{bmatrix} h_{t-1} \\ x_t \end{bmatrix}$$

$$i_t = \sigma(W_i \cdot X + b_i); f_t = \sigma(W_f \cdot X + b_f); o_t = \sigma(W_o \cdot X + b_o)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c \cdot X + b_c)$$

$$h_t = o_t \odot \tanh(c_t),$$

where each of W_i, W_f, W_o are weight matrices that are learned during training in addition to their associated biases, b_i, b_f, b_o . These variables parameterize the transformations of the input, forget and output gates respectively. σ denotes the sigmoid function and $\odot$ is element-wise multiplication. x_t includes the inputs of the LSTM cell unit, i.e., the word embedding vectors w_t ($t = t'$ th word) taken from a sentence $S = \{w_1, w_2, \dots, w_n\}$ and the aspect term embedding v_a . Finally, h_{t-1} is the hidden-state of the previous cell; h_t is the output vector of hidden states; and the final hidden state is the input to the final linear layer before the *softmax* function can output the sentiment probabilities. Refer to the Appendix for the definition of the *softmax* function.

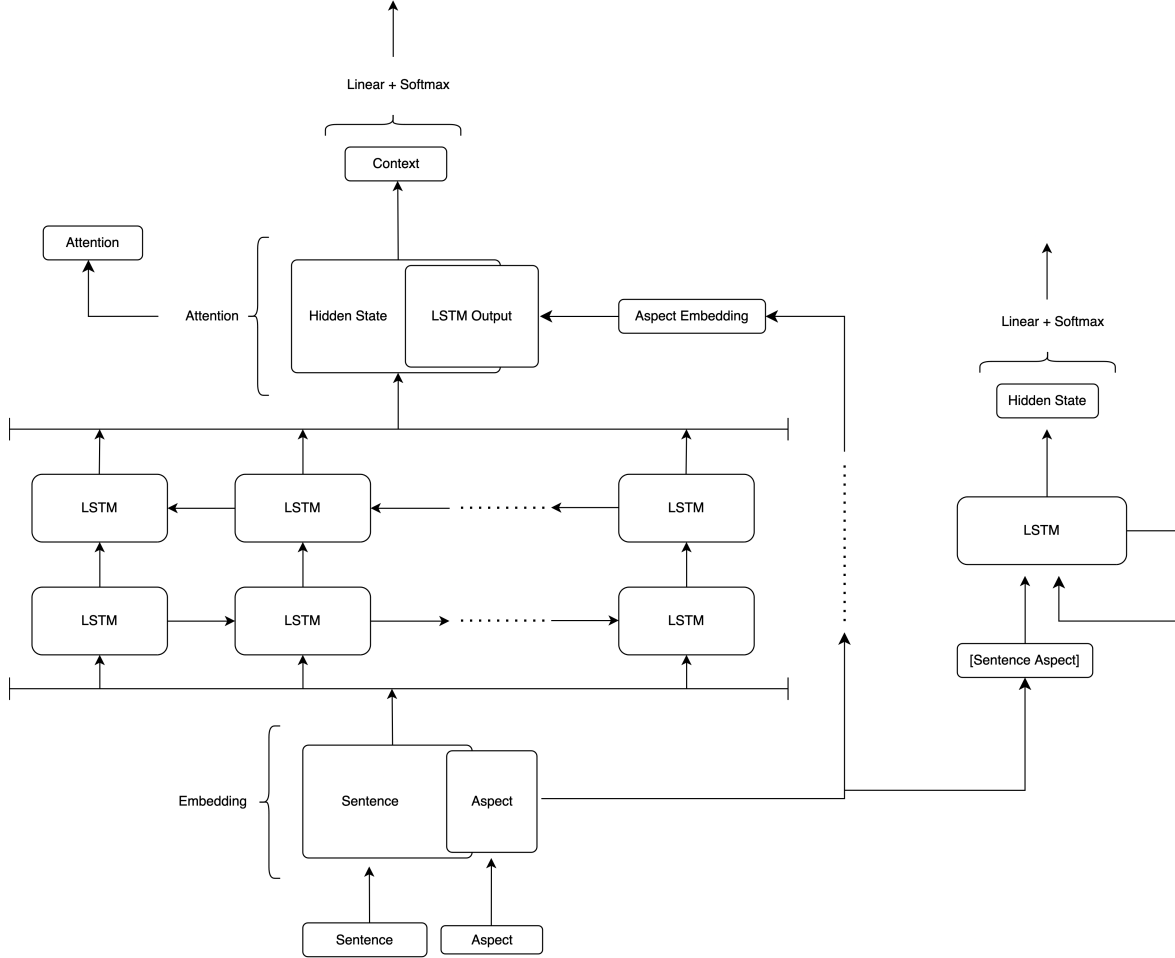


Figure 1. LSTM sentiment classifier model overview.

Turning now to the attention function. There are two forms of attention in our model - *simple attention* and *aspect attention*. Firstly, *simple attention* refers to the dot-product attention function introduced by Luong et al. in *Effective Approaches to Attention-based Neural Machine Translation* [4]. We refer to this form as *simple attention* as it is the simplest approach for computing an attention weighted output, i.e., a *context* vector. This choice was made by design in order to compare the simplest possible approach to attention with the more “*complex*” approach utilized in the *aspect attention* function which we describe now. Following the notation used in [3], we have the matrix H which contains the output states of the LSTM; the aspect embedding v_a ; and the ones vector e_n ; where,

$$\begin{aligned} M &= \tanh\left(\begin{bmatrix} W_h H \\ W_v v_a \otimes e_n \end{bmatrix}\right) \\ \alpha &= \text{softmax}(w^T M) \\ r &= H\alpha^T. \end{aligned}$$

The attention weights α are used to compute the attention weighted hidden representation of the input sentence given the aspect, i.e., r . In our case, H already contains information about the aspect given that H is produced from *both* the input sentence and the associated aspect embeddings. Complementary to this, we have in M , the repeated concatenation of the aspect embedding v_a for as many words as there are in the sentence, i.e., $W_v v_a \otimes e_n$. The weights W_h, W_v are learned during training as are those for the transform $w^T M$. Finally, the context h^* is computed by

$$h^* = \tanh(W_p r + W_x h_n)$$

where, h_n is the final hidden-state of the LSTM, and again, the weights W_p, W_x are learned during training. As in all other cases, the resultant polarity distribution is produced via a linearization layer fed to the *softmax* function.

Experiments

Dataset Description

We use a sub-set of the MAMS challenge dataset introduced by Jiang et al. [2]. Recorded in Table. 1 are the statistics of the dataset - the number of instances in each of the Train, Validation and Test set with the distribution of the sentence polarities respectively.

Table 1: Dataset Statistics

Dataset	Positive	Negative	Neutral	Total
Train	1929	2048	3077	7090
Validation	241	259	388	888
Test	245	263	393	901
Total	2415	2570	3858	8879

As described in [2], “All sentences in MAMS contains multiple aspects with different sentiment polarities.” There are eight different aspect categories: *food, service, staff, price, ambience, menu, place and miscellaneous*.

Experiment Setup

The method we use to test our experimental hypotheses is a case by case comparative analysis of the performance of each of the network configurations related to the hypothesis under consideration. Specifically, we compare the classification performance of the Null Model (Model 0) with the aspect integrated model (Model 1) to test EH1; we compare the classification performance of Model 1 with the *simple attention* model (Model 2) to test EH2; and, finally, we compare the classification performance of Model 2 with the *aspect attention* model (Model 3) to test EH3. We take a balanced approach when evaluating performance. By this we mean that both *accuracy* and *f1* metrics are evaluated on the Test dataset and considered together to either confirm or disconfirm the experimental hypotheses. See the Appendix for definitions of these metrics.

An important consideration in testing our hypotheses is that of confounding. How will we know what variable we can attribute an effect to if we do not have strict controls for our experiments? With this in mind we adopt the requirement when testing the hypotheses that all other variables that do not directly relate to the hypothesis are held constant. In particular, we vary only one network parameter at a time and keep training hyperparameters (i.e., epochs, batch size, learning rate etc.) constant. Specifically, we use the following parameter settings:

- Embedding Dimension: 128
- Hidden Dimension: 256
- Number of LSTM layers: 2

Furthermore, the LSTM layers are bi-directional for each model configuration and the outputs thereof are unregularized. In training the network we utilize the *Adam* optimizer with a learning rate of 0.005; set the default batch size to 10; and train for 10 epochs. Given the nature of the classification task, we employ the *Cross Entropy Loss* as our criterion function (refer to Appendix). Finally, in keeping with our stated goal - to evaluate the effect of differing network architectures as opposed to achieving strictly maximal performance - we utilize learned word embeddings in place of their pretrained counterparts (e.g., Glove etc.).

In order to train the neural network model the data must be in the appropriate form. To meet this requirement we design a data preprocessing pipeline which:

- removes (i.e., expands) contractions in the sentences;
- tokenizes, pads and encodes the sentences and associated aspects;
- encodes polarities (i.e., one-hot vector representations);
- defines a vocabulary (i.e., word-to-index representations);
- and transforms and batches the data into the required tensor format.

A final important point is that the resultant vocabulary (6380 unique tokens) is *unaware* of the Validation and Test data, i.e., a word representation encountered in the Validation or Test data that is not found in the Training data is designated as *unknown*. The reason for this approach is to mitigate the problem of data leakage, i.e., information from the Training data making it's way into the Validation or Test data.

Results

Quantitative Results

The results of our experimental analysis are as follows:

- **Experimental Hypothesis 1:** Integration of aspect information will improve performance over a baseline Null Model that is *unaware* of the aspect - **confirmed**.

- **Experimental Hypothesis 2:** The incorporation of a *simple attention* mechanism to an aspect integrated model will improve classification performance - **confirmed**.
- **Experimental Hypothesis 3:** A model with an aspect integrated attention mechanism, i.e., *aspect attention*, will improve classification performance over the *simple attention* model - **not confirmed**.

Presented in Table. 2 are the quantitative results of the experiment wherein we see that Model 2 is optimal in both accuracy and f1 metrics evaluated on the Test dataset.

Table 2: Experiment Results

Model	Accuracy %	f1
Model 0	43.66	0.2
Model 1	63.44	0.61
Model 2	65.88	0.64
Model 3	65.11	0.62

Recalling that Model 2 is the aspect integrated bi-directional LSTM with *simple attention* and that Model 3 is the aspect integrated bi-directional LSTM with *aspect attention* we notice why the result is surprising. It does not appear, under these experimental conditions, that the more involved attention mechanism produces superior classification performance. Importantly, we observe that the relative difference in accuracy between Model 2 and Model 3 is small, but the relative difference is larger for the f1 metric which is the more robust of the two. We note that there is the possibility that this result is a statistical artefact. It is for this reason that we have conducted two separate ablation studies that are designed to answer questions about the effect of modulating network complexity on sentiment classification performance. The questions are as follows:

- **Question 1:** How does increasing network complexity of the *simple attention* and *aspect attention* models affect (contribute to) performance?
- **Question 2:** How does constraining network complexity (i.e., removing components and/or reducing complexity) affect the optimal attention model?

In this instance, we define *complexity* as the size/density, and hence computational *computational complexity*, of model parameters. Specifically, in **Ablation Study 1.**, we record the effect on comparative performance of Model 2 and Model 3 of increasing the number of recurrent layers of the LSTM (i.e., *Layers* : $2 \rightarrow 4$); increasing the dimensionality of the hidden-state layers of the LSTM (i.e., *HiddenDimension* : $256 \rightarrow 512$); and increasing the dimensionality of the embedding layer (i.e., *EmbeddingDimension* : $128 \rightarrow 256$). We again isolate the variable we are modulating so as not to confound it’s effect. The results of **Ablation Study 1.** are presented in Table. 3, where we see that increasing the dimensionality of the embedding layer in Model 2 is the only parameterization that has a positive effect on accuracy but it is at the expense of a decrease in f1.

Table 3: Ablation Study 1. Results

Model	N Layers	Hidden Dim.	Embedding Dim.	Accuracy %	f1
Model 2	2	256	128	65.88	0.64
Model 3	2	256	128	65.11	0.62
Model 2	4	256	128	48.22	0.31
Model 3	4	256	128	59.44	0.44
Model 2	2	512	128	61.33	0.54
Model 3	2	512	128	40	0.28
Model 2	2	256	256	66.11	0.63
Model 3	2	256	256	62.22	0.6

So we see that Model 2 parameterized with 2 LSTM layers, hidden-state dimension 256, and embedding dimension 128 is the optimal model under these conditions (i.e., the *same* model as identified in Table 2). In **Ablation Study 2.**, we in turn, constrain the model to a single recurrent layer; constrain the model to a uni-directional recurrent layer; and finally, regularize the model via drop out layer integration. We utilize the optimal model found in our previous experiments as the baseline from which to conduct the ablation (i.e., Model 2). The results of **Ablation Study 2.** are presented in Table 4., where we can see that, in terms of constraining network complexity, the overall effect is negative toward accuracy and f1 score but the absolute decline from the completely unconstrained model to the completely constrained model is surprisingly marginal (i.e., absolute decline of approximately 2.33% in accuracy and absolute decline of approximately 0.04 in f1).

Table 4: Ablation Study 2. Results

Model	N Layers	Bidirectional	Dropout Prob.	Accuracy %	f1
Model 2	2	True	0	65.88	0.64
Model 2	1	True	0	64.44	0.62
Model 2	1	False	0	65.33	0.63
Model 2	1	False	0.5	63.55	0.6

The results of our ablations enable us to answer the questions for which they were designed. In summary:

- **Question 1:** How does increasing network complexity of the *simple attention* and *aspect attention* models affect (contribute to) performance? **Increasing network complexity contributes negatively to performance.**
- **Question 2:** How does constraining network complexity (i.e., removing components and/or reducing complexity) affect the optimal attention model? **Constraining network complexity has a surprisingly marginal effect on the optimal attention model.**

The conclusion of this section is that we are able to confirm the result of our initial experiment which identified Model 2 as optimal. However, as described earlier, we have kept the training hyperparameters constant as a form of control. By tuning the training hyperparameters via an *ad hoc* approach, we were able to identify that Model 2 remained the operative model on the basis of classification accuracy and f1 score on the Test dataset. In particular, with a learning rate of 0.0005 and training for 8 epochs, Model 2 achieved an accuracy of 67.77% and f1 score of 0.65 when evaluated on the Test dataset.

Qualitative Results

Presented in Figure 2. we see a visualization of the attention weights for a sentence taken from the Test dataset, namely, “*the food was good but it is not worth the wait or the lousy service*”. The sentence contains two aspects, the first is *food*, the second is *service*. Associated with these aspect terms are the polarities positive and negative respectively. The visualization shows that the model has correctly identified and focused it’s “*attention*” on the important part of the sentence for the *food* aspect. However, for the *service* aspect, the model has it’s “*attention*” only marginally focused on the important part of the sentence whilst the remainder of it’s focus is directed toward an irrelevant part of the sentence. Clearly, in this instance, the model has both relatively good performance and at the same time relatively bad performance.

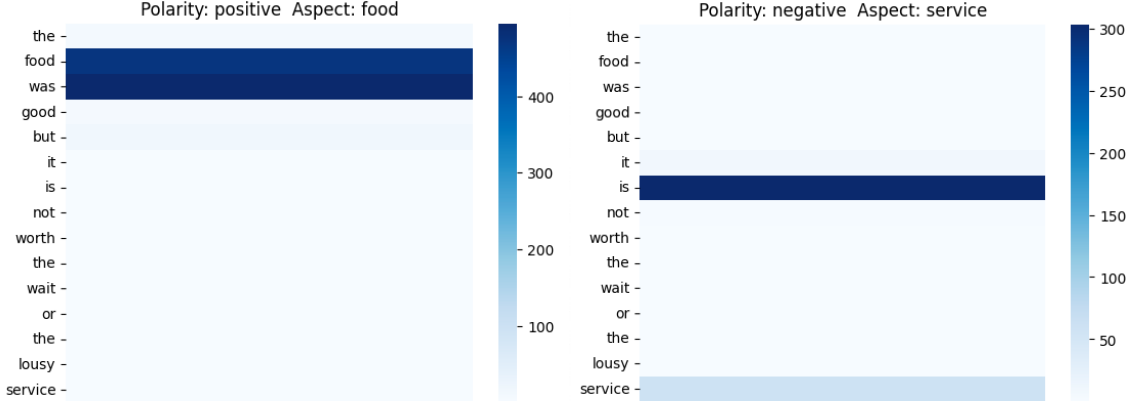


Figure 2. Attention weight visualization (scaled axis).

Conclusion

Recalling that our aim was to formulate several experimental hypotheses about a specific type of sentiment analysis problem (Multi-Aspect-Multi-Sentiment) and propose methods for their evaluation. We have shown the quantitative results lead to the confirmation of EH1 and EH2 but to the disconfirmation of EH3. Furthermore, in the previous sections we presented the quantitative results of several investigations that demonstrate the optimality, under our experimental conditions, of the aspect integrated bi-directional LSTM with *simple attention*, i.e., Model 2. This evidence provides weight against the objection highlighted earlier that this conclusion may be the result of a statistical artefact. With this consideration in mind, it appears that information about the aspect term can only produce better performance to a point - after which diminishing returns are encountered. Recalling also our goal, namely, to determine the effects of differing network architectures as opposed to achieving strictly maximal performance. While this goal was appropriate under the considerations of this investigation, it is also the principal limitation of our results. Specifically, if the goal were to be modified to be affirmative in achieving maximal performance, then a more expansive investigation of the model architecture and its parameterizations would likely produce superior performance and consequently more exciting findings. Additionally, perhaps under that circumstance, the simpler model would not be optimal. With this limitation in mind, we conclude with our main and unexpected finding, that, in this instance, the “*simple*” model is superior to the “*complex*” model.

Appendix

Model description reference

Model	Description
Model 0	aspect <i>unaware</i> Null Model
Model 1	aspect integrated bi-directional LSTM
Model 2	aspect integrated bi-directional LSTM with <i>simple attention</i>
Model 3	aspect integrated bi-directional LSTM with <i>aspect attention</i>

Definitions

$$\text{softmax}(\mathbf{z}) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

$TP = \text{TruePositive}$; $TN = \text{TrueNegative}$; $FP = \text{FalsePositive}$; $FN = \text{FalseNegative}$

$$\begin{aligned} \text{precision} &= \frac{TP}{TP + FP} \\ \text{recall} &= \frac{TP}{TP + FN} \\ \text{accuracy} &= \frac{TP + TN}{TP + FN + TN + FP} \\ f1 &= \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \end{aligned}$$

$$\text{Cross Entropy Loss} = - \sum_i \sum_j y_i^j \log(\hat{y}_j^i)$$

References

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